

Kazakh-British Technical University

Information communication technology

Laboratory work № 2

Normalization

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21.09.2026

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Task 1.

The table is not 1NF, because the “Seat number” attribute contains a not-atomic value (a least of seats: “12A, 12B”). In 1NF all values stored in the relation are single-valued and atomic.

The table violates the 2NF because there are still partial dependencies.

For Example:

- 1) `booking_id` → `passenger_full_name`
- 2) `flight_number` → `departure_airport_name`

The attributes depend not on the entire key, but only on a part of it.

The table violates 3NF because there are transitive dependencies:

For Example:

- 1) `flight_number` → `airline_name`
- 2) `departure_airport_name` → `departure_city`
- 3) `arrival_airport_name` → `arrival_city`

Non-key attributes depend on other non-key attributes.

Task 2.

We are removing the repeating group `seat\_numbers`.

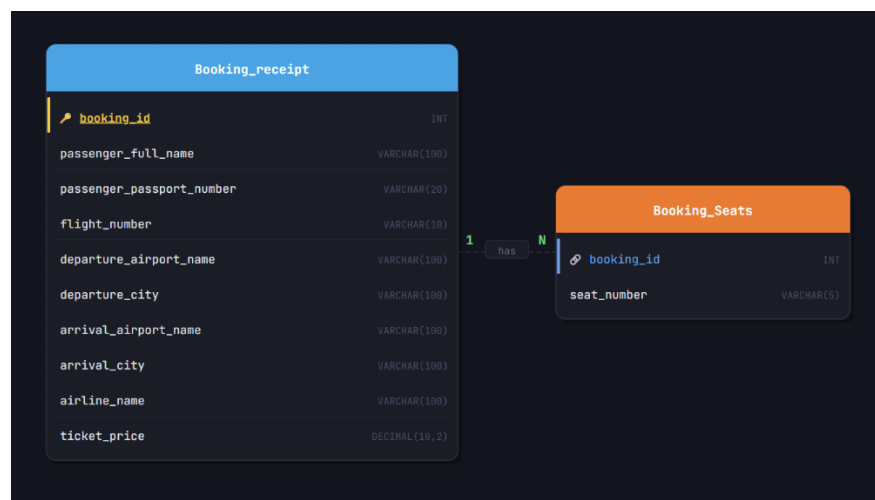


Figure 1. 1NF Table

We decompose the table into entities, eliminating partial dependencies.



Figure 2. 2NF Table

Eliminating transitive dependencies.



Figure 3. 3NF Table

Task 3.

Initial dependencies:

1. `booking_id` -> `passenger_full_name`
2. `booking_id` -> `passenger_passport_number`
3. `booking_id` -> `flight_number`
4. `booking_id` -> `ticket_price`
5. `flight_number` -> `airline_name`
6. `flight_number` -> `departure_airport_name`
7. `flight_number` -> `arrival_airport_name`
8. `departure_airport_name` -> `departure_city`
9. `arrival_airport_name` -> `arrival_city`

After normalization:

1. booking_id -> passenger_id, flight_id, ticket_price
2. passenger_id -> passenger_full_name, passenger_passport_number
3. flight_id -> airline_id, departure_airport_id, arrival_airport_id
4. airline_id -> airline_name
5. airport_id -> airport_name, city

Task 4.

The primary key of the table `Flights` is flight_id.

Functional dependencies:

- 1) flight_id -> airline_id, departure_airport_id, arrival_airport_id

Additional:

- 1) airline_id -> airline_name, city

The `Flights` table does not contain attributes such as airline_name or city. All non-key attributes depend solely on the primary key. So, there are no transitive dependencies within the table.

Task 5.

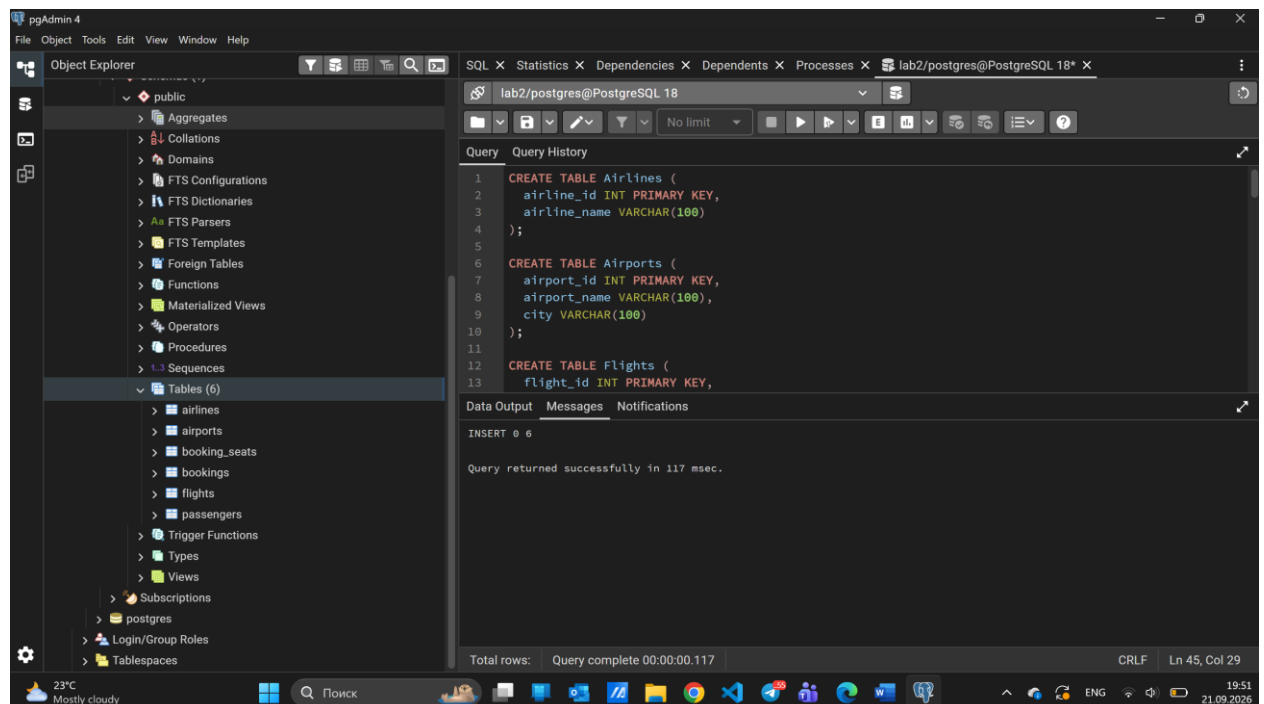


Figure 4. Result of the code

Code: [*Appendix*](#)

DDL

Code:

```
CREATE TABLE Airline_info (  
    airline_id SERIAL PRIMARY KEY,  
    airline_code VARCHAR(30),  
    airline_name VARCHAR(50),  
    airline_country VARCHAR(50),  
    created_at TIMESTAMP,  
    updated_at TIMESTAMP,  
    info VARCHAR(50)  
);  
  
CREATE TABLE Airport (  
    airport_id SERIAL PRIMARY KEY,  
    airport_name VARCHAR(50),  
    country VARCHAR(50),  
    state VARCHAR(50),  
    city VARCHAR(50),  
    created_at TIMESTAMP,  
    updated_at TIMESTAMP  
);  
  
CREATE TABLE Baggage (  
    baggage_id SERIAL PRIMARY KEY,  
    weight_in_kg DECIMAL(4,2),  
    created_at TIMESTAMP,  
    updated_at TIMESTAMP,  
    booking_id INT  
);  
  
CREATE TABLE Baggage_check (  
    baggage_check_id SERIAL PRIMARY KEY,  
    check_result VARCHAR(50),  
    created_at TIMESTAMP,  
    updated_at TIMESTAMP,  
    booking_id INT,  
    passenger_id INT  
);
```

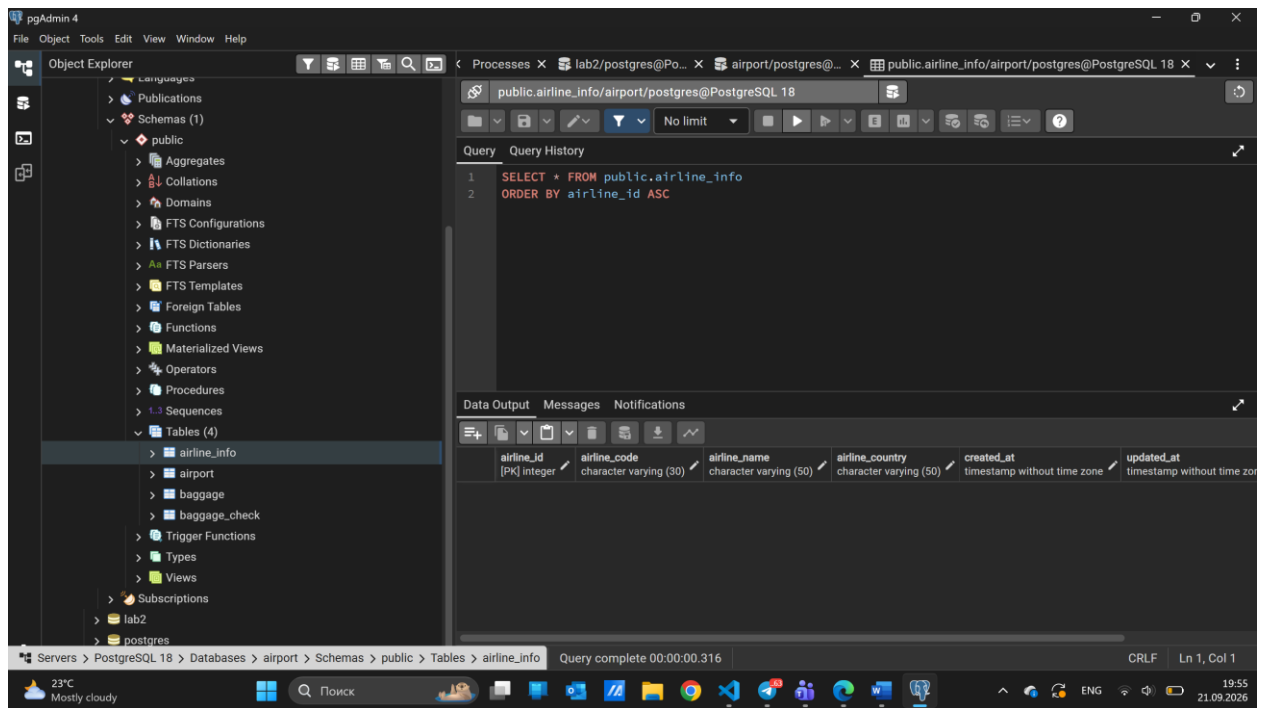


Figure 5. Result of the code

APPENDIX

```
CREATE TABLE Airlines (  
    airline_id INT PRIMARY KEY,  
    airline_name VARCHAR(100)  
);  
  
CREATE TABLE Airports (  
    airport_id INT PRIMARY KEY,  
    airport_name VARCHAR(100),  
    city VARCHAR(100)  
);  
  
CREATE TABLE Flights (  
    flight_id INT PRIMARY KEY,  
    airline_id INT,  
    departure_airport_id INT,  
    arrival_airport_id INT,  
    FOREIGN KEY (airline_id) REFERENCES Airlines(airline_id),  
    FOREIGN KEY (departure_airport_id) REFERENCES Airports(airport_id),  
    FOREIGN KEY (arrival_airport_id) REFERENCES Airports(airport_id)  
);  
  
CREATE TABLE Passengers (  
    passenger_id INT PRIMARY KEY,  
    passenger_full_name VARCHAR(100),  
    passenger_passport_number VARCHAR(50)  
);  
  
CREATE TABLE Bookings (  
    booking_id INT PRIMARY KEY,  
    passenger_id INT,  
    flight_id INT,  
    ticket_price DECIMAL(10,2),  
    FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id),  
    FOREIGN KEY (flight_id) REFERENCES Flights(flight_id)  
);  
  
CREATE TABLE Booking_Seats (  
    booking_id INT,  
    seat_number VARCHAR(5),  
    PRIMARY KEY (booking_id, seat_number),  
    FOREIGN KEY (booking_id) REFERENCES Bookings(booking_id)  
);  
  
-- =====  
  
INSERT INTO Airlines VALUES  
(1, 'Air Astana'),  
(2, 'Lufthansa'),  
(3, 'Turkish Airlines');  
  
INSERT INTO Airports VALUES  
(1, 'Almaty Airport', 'Almaty'),
```

```
(2, 'Astana Airport', 'Astana'),  
(3, 'Frankfurt Airport', 'Frankfurt'),  
(4, 'Istanbul Airport', 'Istanbul');
```

```
INSERT INTO Flights VALUES
```

```
(1, 1, 1, 2),  
(2, 2, 3, 1),  
(3, 3, 4, 1),  
(4, 1, 2, 1),  
(5, 2, 1, 3);
```

```
INSERT INTO Passengers VALUES
```

```
(1, 'Ivan Ivanov', 'P10001'),  
(2, 'Anna Petrova', 'P10002'),  
(3, 'John Smith', 'P10003'),  
(4, 'Ali Khan', 'P10004'),  
(5, 'Maria Garcia', 'P10005');
```

```
INSERT INTO Bookings VALUES
```

```
(1, 1, 1, 500.00),  
(2, 2, 2, 650.00),  
(3, 3, 3, 550.00),  
(4, 4, 4, 300.00),  
(5, 5, 5, 700.00);
```

```
INSERT INTO Booking_Seats VALUES
```

```
(1, '12A'),  
(1, '12B'),  
(2, '14C'),  
(3, '10A'),  
(4, '8B'),  
(5, '16D');
```